



Unit 2: Population and Migration Patterns and Processes

2026–2027

Population Distribution & Density Methods

Physical factors: climate, landforms, water bodies

Human factors: culture, economics, history, politics

Scale matters: factors vary by local, regional, or global analysis

Three density calculation methods:

- **Arithmetic:** total population ÷ total land area (most common, simplest)
- **Physiological:** population ÷ arable land only (shows pressure on farmable land)
- **Agricultural:** population ÷ agricultural land (includes pasture)

Key insight: Method chosen reveals different information about land pressure and resource strain.

Aging Populations & Consequences

Causes: low birth rates + high life expectancy = aging population

Dependency ratio: (young + old dependents) ÷ working-age population

- High ratio = more non-workers per worker
- Strains pensions, health care, social services

Consequences:

- **Economic:** labor shortages, pension/health-care costs, reduced consumer demand
- **Social:** isolation, caregiver burden, intergenerational tension
- **Political:** policy shifts toward elderly, immigration debates

Distribution, Density & Environment

Effects on society: population distribution and density affect political, economic, and social processes—especially service provision (medical care, education, infrastructure).

Effects on environment: carrying capacity = maximum population an environment can sustain.

- High density → resource depletion, pollution, waste management strain
- Low density → underutilized infrastructure, sprawl
- Uneven distribution → regional inequality in resource access

Test focus: link density patterns to real-world service gaps and environmental stress.

Population Composition & Pyramids

Age structure & sex ratio vary across regions and scales; can be mapped and analyzed.

Population pyramids show:

- Age cohorts (horizontal bars)
- Sex distribution (left/right sides)
- Growth or decline patterns
- Future market demand for goods/services

Pyramid shapes indicate:

- **Expansive (wide base):** high fertility, young population, rapid growth
- **Constrictive (narrow base):** low fertility, aging population, slow/negative growth
- **Stationary (rectangular):** stable population

Population Dynamics: Growth & Decline

Three demographic factors determine growth/decline:

- Fertility (births)
- Mortality (deaths)
- Migration (in/out)

Key measures:

- **Rate of Natural Increase (RNI):** (births – deaths) ÷ population × 1000
- **Population-doubling time:** 70 ÷ RNI (years to double at current rate)

Influences on rates: social, cultural, political, and economic factors shape fertility, mortality, and migration patterns.

Demographic Transition Model (DTM)

Four stages explain population change over time:

Stage	Birth Rate	Death Rate	Growth
1 (Pre-transition)	High	High	Low
2 (Early transition)	High	Falling	Rapid
3 (Late transition)	Falling	Low	Slowing
4 (Post-transition)	Low	Low	Low/stable

Epidemiological transition: explains causes of changing death rates (disease patterns shift with development).

Malthusian Theory & Critiques

Malthus (1798): population grows exponentially; food supply grows arithmetically → inevitable crisis.

Used to: analyze population change and its consequences.

Major critiques:

- Ignored technological innovation in agriculture
- Underestimated human adaptability
- Did not account for voluntary fertility decline
- Predictions have not materialized in developed nations

Still relevant for: discussing resource limits, carrying capacity, and sustainability debates.

Migration: Types & Causes

Push factors: conditions that encourage leaving (poverty, conflict, poor services, environmental hazard)

Pull factors: conditions that attract arrival (jobs, safety, education, better climate)

Intervening opportunities/obstacles: distance, cost, legal barriers, family ties, language

Can be: cultural, demographic, economic, environmental, or political

Voluntary migration types: transnational, transhumance, internal, chain, step, guest worker, rural-to-urban

Forced migration types: slavery, refugee crises, internally displaced persons (IDPs), asylum seekers

Population Policies & Women's Roles

Population policies:

- **Pronatalist:** encourage population growth (incentives, family benefits)
- **Antinatalist:** discourage growth (one-child policies, contraception access)
- **Immigration policies:** control in/out migration

Women's demographic impact:

- Education, employment, health care access reduce fertility
- Contraception availability lowers birth rates
- Changing social/economic/political roles influence fertility, mortality, migration
- **Ravenstein's laws of migration:** migration patterns linked to female mobility and opportunity